

COVID-19 Epidemiological Dashboard

A Public Health Data Analytics Project

Prepared by

Oumar Mahamat Ahmat

MPH in Epidemiology

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Executive Summary

This project presents the development of an interactive COVID-19 Epidemiological Dashboard designed to analyze global COVID-19 surveillance data and support evidence-based public health decision-making. The primary objective was to demonstrate practical skills in epidemiological data analysis by transforming raw surveillance data into meaningful indicators and interactive visualizations.

The analysis was based on the World Health Organization (WHO) COVID-19 Global Surveillance Dataset, which includes reported cases, deaths, countries, regions, and reporting dates. Data preprocessing and exploratory data analysis (EDA) were performed using Python with libraries including Pandas, NumPy, and Matplotlib. Key epidemiological indicators, such as the Case Fatality Rate (CFR), Growth Rate, and 7-Day Moving Average, were calculated to better understand disease trends over time.

An interactive dashboard was developed using Microsoft Power BI, featuring KPI cards, time-series analysis, geographic maps, regional comparisons, and interactive filters by country and date. The analysis identified substantial differences in COVID-19 burden across countries and WHO regions, revealed multiple epidemic waves during the study period, and estimated an average global Case Fatality Rate (CFR) of approximately 1.21%. This project demonstrates the application of data analytics and business intelligence tools to epidemiology and public health and serves as a portfolio project showcasing skills in Python, Power BI, and health data visualization.

Introduction

The Coronavirus Disease 2019 (COVID-19) pandemic, caused by the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), emerged in late 2019 and rapidly spread across the world, becoming one of the most significant public health emergencies in modern history. In March 2020, the World Health Organization (WHO) declared COVID-19 a global pandemic due to its widespread transmission and substantial impact on health systems, economies, and societies. The pandemic resulted in millions of confirmed cases and deaths worldwide, highlighting the importance of timely disease surveillance and evidence-based decision-making.

Epidemiological data analysis plays a crucial role in understanding the dynamics of infectious diseases. By analyzing surveillance data, public health professionals can monitor disease trends, identify high-risk populations and geographic areas, evaluate the severity of outbreaks, and measure the effectiveness of public health interventions. Key epidemiological indicators, such as confirmed cases, deaths, Case Fatality Rate (CFR), growth rate, and moving averages, provide valuable insights into disease transmission patterns and support informed public health responses.

Interactive dashboards have become essential tools for transforming complex surveillance data into clear and actionable information. By combining data visualization with real-time filtering and trend analysis, dashboards enable policymakers, epidemiologists, researchers, and healthcare managers to quickly identify emerging patterns, compare disease burden across countries and regions, and monitor changes over time. These tools enhance communication, facilitate evidence-based planning, and support strategic decision-making during public health emergencies.

This project demonstrates the application of Python and Microsoft Power BI to analyze the WHO COVID-19 Global Surveillance Dataset and develop an interactive epidemiological dashboard. Through data cleaning, exploratory data analysis, calculation of key epidemiological indicators, and interactive visualization, the project illustrates how modern data analytics techniques can support disease surveillance and strengthen public health decision-making.

Project Objectives

The main objective of this project is to develop an interactive epidemiological dashboard that supports the analysis and visualization of global COVID-19 surveillance data. Specifically, the project aims to:

- Analyze global COVID-19 trends using the World Health Organization (WHO) surveillance dataset.
- Perform data cleaning, preprocessing, and exploratory data analysis (EDA) to improve data quality and identify meaningful patterns.

- Calculate key epidemiological indicators, including the Case Fatality Rate (CFR), Growth Rate, and 7-Day Moving Average.
- Develop an interactive dashboard using Microsoft Power BI to visualize cases, deaths, regional distribution, and temporal trends.
- Enable users to explore COVID-19 data through interactive filters by country and reporting date.
- Compare the burden of COVID-19 across countries and WHO regions using charts, maps, and key performance indicators (KPIs).
- Demonstrate practical skills in Python, Power BI, and epidemiological data analysis through a real-world public health project.
- Showcase the application of data analytics and visualization techniques to support evidence-based public health decision-making

Dataset Description

The dataset used in this project was obtained from the World Health Organization (WHO) COVID-19 Global Surveillance Dataset, which provides official daily reports on the global spread of COVID-19. The dataset contains country-level surveillance information and is intended to support public health monitoring, epidemiological analysis, and evidence-based decision-making.

The dataset consists of daily time-series data, with each record representing the reported COVID-19 situation for a specific country on a given reporting date. It includes information on new and cumulative confirmed cases and deaths, as well as the corresponding WHO region. Prior to analysis, the dataset was cleaned and preprocessed using Python to ensure consistency and improve data quality.

Dataset Source

- ❖ **Organization:** World Health Organization (WHO)
- ❖ **Dataset:** COVID-19 Global Surveillance Dataset
- ❖ **Data Type:** Time-series epidemiological surveillance data
- ❖ **Geographical Coverage:** Global (WHO Member States and territories)

❖ **Time Unit:** Daily reported observations

Key Variables

Variable	Description
Date reported	Date when COVID-19 data were officially reported.
Country	Name of the reporting country or territory.
WHO_region	WHO regional classification (e.g., AFRO, AMRO, EURO, EMRO, SEARO).
New_cases	Number of newly confirmed COVID-19 cases reported on a specific date.
Total_cases	Cumulative number of confirmed COVID-19 cases.
New_deaths	Number of newly reported COVID-19 deaths.
Total_deaths	Cumulative number of COVID-19 deaths.

Methodology

This project followed a structured data analytics workflow consisting of five main phases, from data acquisition to the development of an interactive dashboard. Each phase contributed to transforming raw surveillance data into meaningful epidemiological insights.

Phase 1: Data Collection

The COVID-19 dataset was obtained from the World Health Organization (WHO) Global Surveillance Dataset. The dataset contains daily country-level reports of confirmed COVID-19 cases and deaths, covering multiple WHO regions. It served as the primary source of epidemiological data for this project.

Phase 2: Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed using Python to ensure the dataset was accurate, consistent, and ready for analysis. The following tasks were completed:

- Imported the dataset into Jupyter Notebook using Pandas.
- Converted the reporting date into a datetime format.
- Checked for missing values and data inconsistencies.

- Verified and corrected data types.
- Renamed variables where necessary for consistency.
- Prepared the cleaned dataset for exploratory analysis and dashboard development.

Phase 3: Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify patterns and summarize the characteristics of the dataset. The analysis included:

- Daily trends of confirmed COVID-19 cases.
- Comparison of cumulative cases and deaths across countries.
- Distribution of cases by WHO region.
- Summary statistics for key variables.
- Correlation analysis between epidemiological indicators.

The findings from this phase guided the selection of visualizations and indicators used in the dashboard.

Phase 4: Epidemiological Indicators

Several epidemiological indicators were calculated to provide deeper insights into the progression and severity of the pandemic.

The indicators included:

- Case Fatality Rate (CFR): The proportion of reported deaths among confirmed cases.
- Growth Rate: The percentage change in cumulative confirmed cases over time.
- 7-Day Moving Average: A smoothed trend of daily reported cases to reduce daily fluctuations and better identify epidemic waves.

These indicators support the interpretation of disease trends and facilitate comparisons across countries and time periods.

Phase 5: Dashboard Development

An interactive dashboard was developed using Microsoft Power BI to present the analysis in a user-friendly format. The dashboard includes:

- Key Performance Indicator (KPI) cards for Total Cases, Total Deaths, and Average CFR.
- Time-series visualization of daily COVID-19 cases.
- Geographic map showing the global distribution of cases.
- Regional comparisons based on WHO regions.
- Interactive filters for country and reporting date.

An "About this Project" page describing the project, data source, tools, and objectives.

The dashboard enables users to explore COVID-19 trends dynamically and supports evidence-based public health decision-making

7/ Results

The analysis of the WHO COVID-19 Global Surveillance Dataset provided valuable insights into the global progression of the COVID-19 pandemic. The interactive dashboard enabled users to explore epidemiological trends across countries and WHO regions through dynamic visualizations and key performance indicators.

7.1 Dashboard Overview

The dashboard presents a comprehensive overview of the pandemic through interactive KPI cards, charts, maps, and filters. Users can explore the data by country and reporting date to understand better temporal and geographical variations in COVID-19 cases and deaths.

The main dashboard components include:

- Total confirmed COVID-19 cases.
- Total reported deaths.
- Average Case Fatality Rate (CFR).
- Daily trend of new confirmed cases.
- Geographic distribution of cases.
- WHO regional comparisons.
- Interactive country and date filters.

7.2 Key Findings

The analysis produced the following key findings:

The COVID-19 pandemic affected all WHO regions, although the magnitude of transmission varied considerably between countries.

Multiple epidemic waves were observed during the reporting period, as reflected in the daily trend in new cases.

Countries with the highest cumulative case counts contributed a substantial proportion of the global disease burden.

The calculated Case Fatality Rate (CFR) indicated differences in disease severity and reporting patterns across countries.

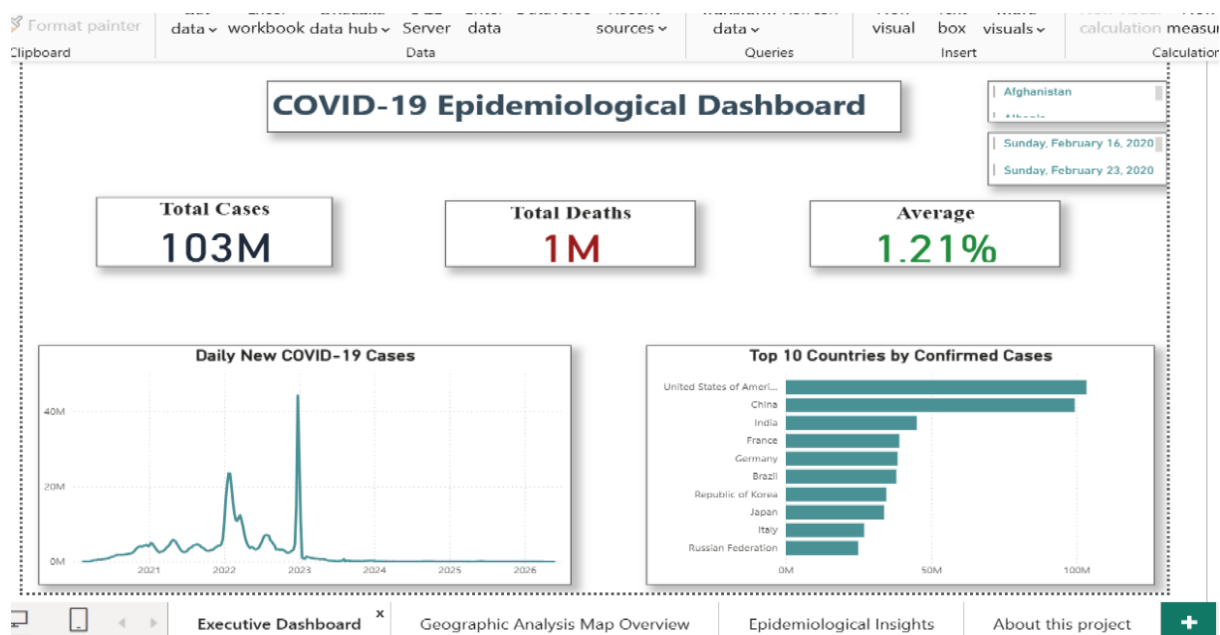
Geographic visualization highlighted the global distribution of COVID-19 and identified regions with the highest cumulative case counts.

7.3 Dashboard Visualizations

The Power BI dashboard consists of four pages:

Page 1 – COVID-19 Epidemiological Dashboard

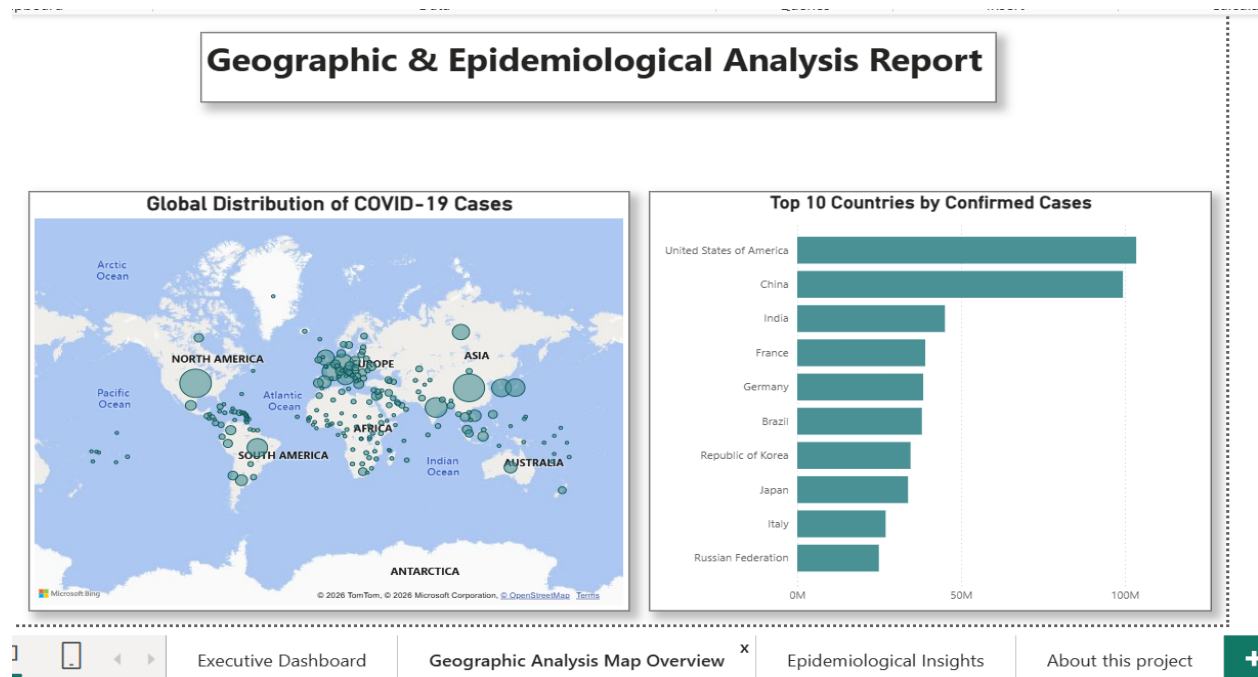
Provides an executive summary of the pandemic using KPI cards, trend charts, and country comparisons.



Page 2 – Epidemiological Insights

Presents advanced epidemiological analyses, including WHO regional comparisons, trends, and case fatality rate indicators.

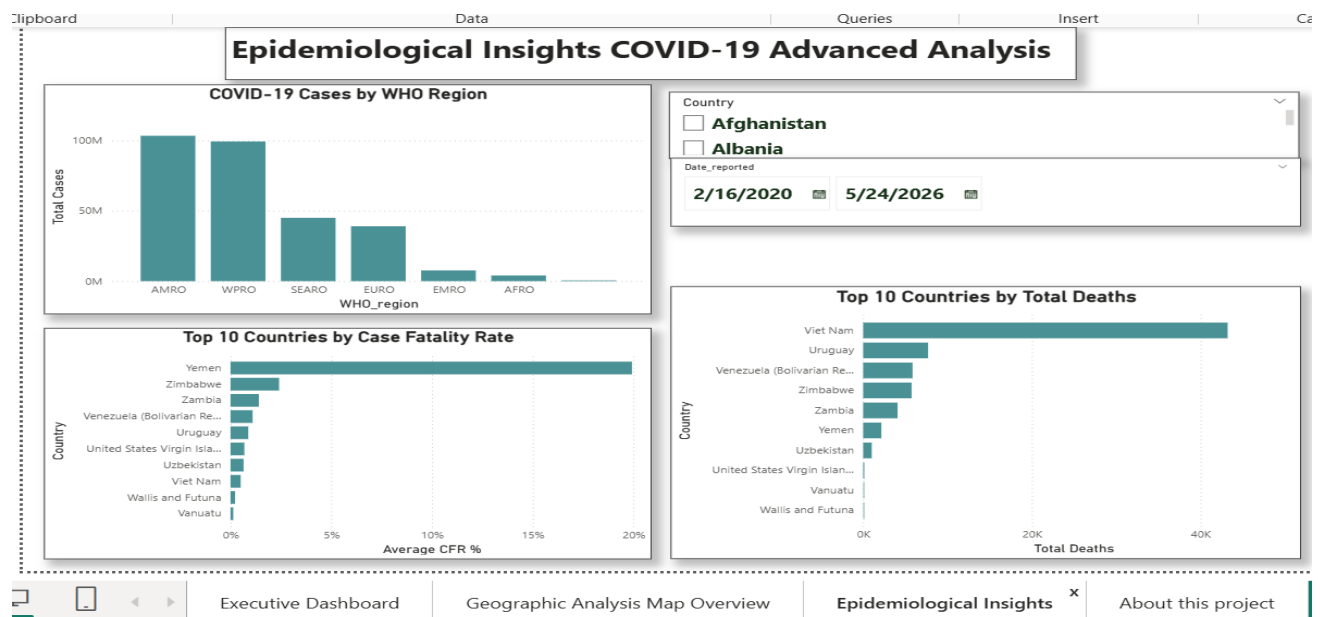
Figure 2: Epidemiological Insights Screenshot



Page 3 – Geographic Analysis

Displays the global geographic distribution of COVID-19 cases using an interactive map.

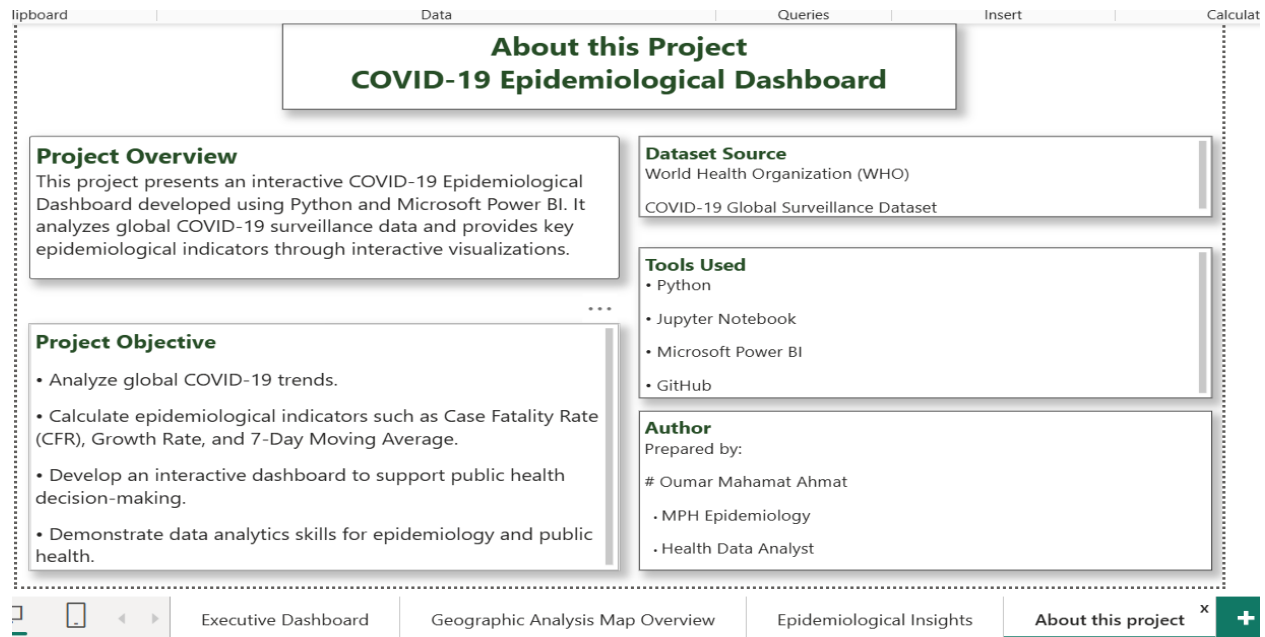
Figure 3: Geographic Analysis Screenshot



Page 4 – About this Project

Provides background information about the project, including its objectives, data source, analytical tools, and dashboard features.

Figure 4: About this Project Screenshot



7.4 Overall Outcome

The project successfully transformed raw COVID-19 surveillance data into an interactive and informative dashboard. By integrating Python for data preprocessing and Microsoft Power BI for visualization, the dashboard enables efficient exploration of epidemiological patterns and supports evidence-based public health decision-making. It also demonstrates practical competencies in data cleaning, exploratory data analysis, epidemiological indicator calculation, and interactive dashboard development.

Key Findings

The analysis of the WHO COVID-19 Global Surveillance Dataset revealed several important epidemiological findings:

- The COVID-19 pandemic affected all WHO regions, with substantial differences in the number of reported cases and deaths across countries.
- Global cumulative confirmed cases exceeded 100 million during the study period, demonstrating the widespread impact of the pandemic.
- The estimated global Case Fatality Rate (CFR) was approximately 1.21%, indicating variation in disease severity and reporting across countries.
- The daily case trend revealed multiple epidemic waves, reflecting periods of increasing and decreasing transmission over time.
- Geographic analysis showed that the burden of COVID-19 was unevenly distributed, with some countries reporting significantly higher cumulative case counts than others.
- WHO regional comparisons highlighted considerable differences in the progression and impact of the pandemic across regions.
- Interactive visualizations enabled users to explore trends by country and reporting date, facilitating more detailed epidemiological analysis.
- The integration of Python and Microsoft Power BI successfully transformed raw surveillance data into an interactive dashboard that supports evidence-based public health decision-making.

Conclusion

This project successfully demonstrated the application of data analytics and visualization techniques to support epidemiological analysis of the COVID-19 pandemic. Using the World Health Organization (WHO) COVID-19 Global Surveillance Dataset, the project transformed raw surveillance data into meaningful public health information through systematic data cleaning, exploratory data analysis, epidemiological indicator calculation, and interactive visualization.

Python was used to preprocess the data and calculate key epidemiological indicators, while Microsoft Power BI was used to develop an interactive dashboard that allows users to explore COVID-19 trends by country, reporting date, and WHO region. The dashboard provides a clear overview of the global burden of the pandemic through KPI cards, trend analyses, geographic mapping, and regional comparisons.

The project demonstrates the value of integrating epidemiology, data science, and business intelligence tools to improve the interpretation of surveillance data and support evidence-based public health decision-making. It also highlights the importance of interactive dashboards in communicating complex epidemiological information in a clear and accessible manner.

Overall, this project strengthened practical skills in Python, Power BI, data visualization, epidemiological analysis, and public health data analytics, providing a solid portfolio project for future opportunities in epidemiology, health informatics, and public health data analysis.

Recommendations

Based on the findings of this project, the following recommendations are proposed to enhance future epidemiological analyses and improve the functionality of the dashboard:

- Integrate vaccination data to evaluate the relationship between vaccination coverage and COVID-19 trends across countries and regions.
- Include demographic variables, such as age, sex, and population size, to enable more detailed epidemiological analyses and identify vulnerable population groups.

- Incorporate additional public health indicators, including testing rates, hospitalization data, intensive care unit (ICU) admissions, and recovery rates, to provide a more comprehensive assessment of the pandemic.
- Develop predictive models using machine learning to forecast future case trends and support early warning systems for disease surveillance.
- Automate dashboard updates by connecting Power BI to regularly updated data sources, ensuring timely and accurate reporting.
- Expand geographic analysis by including subnational data (e.g., provinces, states, or districts) where available to support localized public health interventions.
- Enhance dashboard interactivity by adding drill-through pages, advanced filtering options, and customizable visualizations for different user groups.
- Apply the same analytical framework to other public health priorities, such as malaria, tuberculosis, HIV/AIDS, and routine immunization surveillance, to support broader public health monitoring

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